Exercise 1 — Step-by-Step Solutions

Optimization-Based Control Methods · Linear Systems and Linear Controllers

Course: Optimization-Based Control Methods (TU Chemnitz, Automatic Control Group, Dr. Adrian Redder) **Covers:** eigenvalue practice, Task 0 (all parts), Task 1 (all parts) **Status:** Tasks 2, 3 and the numerical experiment will be added later

Every step is written out. Section references like "Concepts §5.4" point to `01_Concepts.md`, which explains the theory behind each step. The Python scripts in `python/` verify every number.

Contents

- Part A — Eigenvalue practice
- Part B — Task 0: Controllable and observable subspaces
- Part C — Task 1: Tracking error for a state-feedback controller

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Part A — Eigenvalue practice

Method used throughout (Concepts §2.3): (1) form $\lambda I - A$; (2) take $\det(\lambda I - A)$; (3) find the roots; (4) check $\sum \lambda = \operatorname{tr} A$ and $\prod \lambda = \det A$.

Stability rule (Concepts §4.1): $x_{k+1} = Ax_k$ is asymptotically stable $\iff$ every eigenvalue has $|\lambda| < 1$.

Problem 1

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1. $\lambda I - A = \begin{bmatrix} \lambda - 4 & -1 \\ -2 & \lambda - 3 \end{bmatrix}$.

Step 2. $\det(\lambda I - A) = (\lambda - 4)(\lambda - 3) - (-1)(-2) = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10$.

Shortcut check: $\operatorname{tr} A = 7$ and $\det A = 4 \cdot 3 - 1 \cdot 2 = 10$ give the same polynomial $\lambda^2 - 7\lambda + 10$.

Step 3. Look for two numbers with product 10 and sum -7: these are -2 and -5. So $(\lambda - 2)(\lambda - 5) = 0$.

$$\boxed{\lambda = 2,\ 5}$$

Step 4. $2 + 5 = 7 = \operatorname{tr} A$ ✓, $2 \cdot 5 = 10 = \det A$ ✓.

Stable? No: $|2| > 1$ and $|5| > 1$.

Problem 2

$$A = \begin{bmatrix} 3 & 5 \\ 0 & -1 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda - 3 & -5 \\ 0 & \lambda + 1 \end{bmatrix}$, so $\det = (\lambda - 3)(\lambda + 1) - (-5)(0) = (\lambda - 3)(\lambda + 1)$.

The 0 below the diagonal removes the off-diagonal product. This is why **triangular matrices have their diagonal entries as eigenvalues**.

$$\boxed{\lambda = 3,\ -1}$$

Check: $3 + (-1) = 2 = \operatorname{tr} A$ ✓, $3 \cdot (-1) = -3 = \det A$ ✓. **Stable?** No: $|3| > 1$. ($|-1| = 1$ is also not strictly inside the unit circle.)

Problem 3

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{bmatrix}$, so $\det = \lambda(\lambda + 3) - (-1)(2) = \lambda^2 + 3\lambda + 2$.

Look for two numbers with product 2 and sum 3: these are 1 and 2. So $(\lambda + 1)(\lambda + 2) = 0$.

$$\boxed{\lambda = -1,\ -2}$$

Check: $-1 - 2 = -3 = \operatorname{tr} A$ ✓, $(-1)(-2) = 2 = \det A$ ✓.

Note: in this **controllable canonical form**, the bottom row $[-2,\ -3]$ holds the coefficients of $\lambda^2 + 3\lambda + 2$ with the signs flipped.

Stable? No: $|-2| = 2 > 1$ (and $|-1| = 1$ is on the boundary).

Problem 4

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda - 1 & 2 \\ -2 & \lambda - 1 \end{bmatrix}$, so $\det = (\lambda - 1)^2 - (2)(-2) = (\lambda - 1)^2 + 4$.

Set it to zero: $(\lambda - 1)^2 = -4 \Rightarrow \lambda - 1 = \pm\sqrt{-4} = \pm 2i$.

(Same result with the formula: $\operatorname{tr} A = 2$, $\det A = 1 + 4 = 5$, $\lambda = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm 4i}{2}$.)

$$\boxed{\lambda = 1 \pm 2i}$$

Check: $(1 + 2i) + (1 - 2i) = 2$ ✓, $(1 + 2i)(1 - 2i) = 1 - 4i^2 = 1 + 4 = 5$ ✓. Magnitude: $|\lambda| = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.24$. Complex eigenvalues mean rotation, and $|\lambda| > 1$ means the state spirals outward. **Stable?** No.

Problem 5

$$A = \begin{bmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda - 0.5 & -0.2 \\ -0.1 & \lambda - 0.4 \end{bmatrix}$, so $\det = (\lambda - 0.5)(\lambda - 0.4) - (0.2)(0.1) = \lambda^2 - 0.9\lambda + 0.20 - 0.02 = \lambda^2 - 0.9\lambda + 0.18$.

Quadratic formula: discriminant $= 0.9^2 - 4(0.18) = 0.81 - 0.72 = 0.09$, and $\sqrt{0.09} = 0.3$:

$$\lambda = \frac{0.9 \pm 0.3}{2} \quad \Rightarrow \quad \boxed{\lambda = 0.6,\ 0.3}$$

Check: $0.6 + 0.3 = 0.9$ ✓, $0.6 \cdot 0.3 = 0.18$ ✓. **Stable? Yes**, since $|0.6| < 1$ and $|0.3| < 1$. This is the only stable matrix among the eight problems.

Problem 6

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & -2 & 0 \\ 7 & 5 & 0.5 \end{bmatrix}$$

Step 1. Put λ on the diagonal and flip every sign. The diagonal entries become $\lambda - 1$, $\lambda - (-2) = \lambda + 2$ and $\lambda - 0.5$:

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & 0 & 0 \\ -4 & \lambda + 2 & 0 \\ -7 & -5 & \lambda - 0.5 \end{bmatrix}.$$

Step 2. Expand along row 1, $(\lambda - 1,\ 0,\ 0)$. The two zeros remove their terms:

$$\det(\lambda I - A) = +(\lambda - 1)\det \begin{bmatrix} \lambda + 2 & 0 \\ -5 & \lambda - 0.5 \end{bmatrix} - 0 \cdot (\dots) + 0 \cdot (\dots).$$

The 2×2 determinant is $(\lambda + 2)(\lambda - 0.5) - (0)(-5) = (\lambda + 2)(\lambda - 0.5)$. Therefore

$$\det(\lambda I - A) = (\lambda - 1)(\lambda + 2)(\lambda - 0.5).$$

Step 3. Set each factor to zero:

$$\boxed{\lambda = 1,\ -2,\ 0.5}$$

Step 4. Sum: $1 - 2 + 0.5 = -0.5 = \operatorname{tr} A$ ✓. Product: $1 \cdot (-2) \cdot 0.5 = -1 = \det A$ (the product of the diagonal for a triangular matrix) ✓.

Why the shortcut works: the entries below the diagonal $(4, 7, 5)$ never contributed, because each time they would appear they are multiplied by a 0 from above the diagonal.

Stable? No. $|-2| = 2 > 1$: that mode doubles in size and flips sign every step. $\lambda = 1$ never shrinks either. Only the $\lambda = 0.5$ mode dies out.

Problem 7

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & 0 & 0 \\ 0 & \lambda - 1 & -4 \\ 0 & -2 & \lambda - 3 \end{bmatrix}.$$

Expand along row 1:

$$\det = (\lambda-2)\big[(\lambda-1)(\lambda-3)-(-4)(-2)\big] = (\lambda-2)\big[\lambda^2-4\lambda+3-8\big] = (\lambda-2)(\lambda^2-4\lambda-5).$$

Factor: two numbers with product -5 and sum -4 are -5 and $+1$, so $\lambda^2-4\lambda-5 = (\lambda-5)(\lambda+1)$.

$$\boxed{\lambda = 2,\ 5,\ -1}$$

Check: $2+5-1 = 6 = 2+1+3$ ✓. $2\cdot5\cdot(-1) = -10$, and $\det A = 2\,(1\cdot3-4\cdot2) = 2(-5) = -10$ ✓. **Stable?** No.

Problem 8

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Step 1.

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & 0 & -1 \\ 0 & \lambda - 3 & 0 \\ -1 & 0 & \lambda - 2 \end{bmatrix}.$$

Step 2: choose the row. The middle row $(0,\ \lambda-3,\ 0)$ has two zeros, so only one term survives.

Step 3: sign. By the checkerboard $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$, position $(2,2)$ has sign $+$.

Step 4: expand. Delete row 2 and column 2. What remains is $\begin{bmatrix} \lambda-2 & -1 \\ -1 & \lambda-2 \end{bmatrix}$:

$$\det(\lambda I - A) = +(\lambda - 3)\det\begin{bmatrix} \lambda-2 & -1 \\ -1 & \lambda-2 \end{bmatrix}.$$

Step 5: the 2×2 determinant. $(\lambda - 2)(\lambda - 2) - (-1)(-1) = (\lambda - 2)^2 - 1$. Careful: $(-1)(-1) = +1$, and it is *subtracted*.

Step 6: factor. Use the difference of squares $a^2 - b^2 = (a - b)(a + b)$ with $a = \lambda - 2$, $b = 1$:

$$(\lambda - 2)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

(Or expand: $\lambda^2 - 4\lambda + 4 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$.)

$$\det(\lambda I - A) = (\lambda - 3)^2(\lambda - 1) \quad \Rightarrow \quad \boxed{\lambda = 1,\ 3,\ 3}$$

Step 7: check. $1 + 3 + 3 = 7 = 2 + 3 + 2$ ✓. $1 \cdot 3 \cdot 3 = 9$, and $\det A = 3\,(2 \cdot 2 - 1 \cdot 1) = 9$ (expanding along the middle row) ✓.

Same result expanding along row 1 (signs $+, -, +$):

$$\begin{aligned}\det &= +(\lambda - 2)\det\begin{bmatrix}\lambda - 3 & 0\\ 0 & \lambda - 2\end{bmatrix} - 0 + (-1)\det\begin{bmatrix}0 & \lambda - 3\\ -1 & 0\end{bmatrix}\\ &= (\lambda - 2)(\lambda - 3)(\lambda - 2) + (-1)\big[0 \cdot 0 - (\lambda - 3)(-1)\big]\\ &= (\lambda - 3)(\lambda - 2)^2 - (\lambda - 3) = (\lambda - 3)\big[(\lambda - 2)^2 - 1\big].\end{aligned}$$

It is the same expression, but with more work. That is why choosing the row with the most zeros pays off.

Stable? No: $|3| > 1$, so that mode triples every step.

Summary of Part A

#	Eigenvalues	Stable?
1	$2,\ 5$	no
2	$3,\ -1$	no
3	$-1,\ -2$	no
4	$1 \pm 2i$ ($\lvert\lambda\rvert = \sqrt{5}$)	no
5	$0.6,\ 0.3$	**yes**
6	$1,\ -2,\ 0.5$	no
7	$2,\ 5,\ -1$	no
8	$1,\ 3,\ 3$	no

Part B — Task 0: Controllable and observable subspaces

System:

$$x_{k+1} = \underbrace{\begin{bmatrix}1 & 0 & 0\\ 0 & 2 & 2\\ 0 & 1 & 3\end{bmatrix}}_{A} x_k + \underbrace{\begin{bmatrix}1\\ 0\\ 1\end{bmatrix}}_{B} u_k, \qquad y_k = \underbrace{\begin{bmatrix}1 & 1 & 1\end{bmatrix}}_{C} x_k.$$

Here $n = 3$ states, $m = 1$ input and $p = 1$ output.

B.0 Preparation: eigenvalues and eigenvectors of A

From Concepts §2.7: $\det(\lambda I - A) = (\lambda - 1)^2(\lambda - 4)$, so $\lambda = 1, 1, 4$.

Eigenvector for $\lambda = 4$: solve $(4I - A)v = 0$:

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -1 & 1 \end{bmatrix} v = 0 \;\Rightarrow\; 3v_1 = 0,\; 2v_2 - 2v_3 = 0 \;\Rightarrow\; v = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Check: $A[0,1,1]^\top = [0,\, 2+2,\, 1+3]^\top = [0,4,4]^\top = 4 \cdot [0,1,1]^\top$ ✓.

Eigenvectors for $\lambda = 1$: solve $(I - A)v = 0$:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix} v = 0 \;\Rightarrow\; -v_2 - 2v_3 = 0 \;\Rightarrow\; v_2 = -2v_3,\; v_1 \text{ free.}$$

This gives two independent eigenvectors, $[1,0,0]^\top$ and $[0,-2,1]^\top$, so the eigenspace is a plane. In particular, $[1,-2,1]^\top = [1,0,0]^\top + [0,-2,1]^\top$ is also an eigenvector for $\lambda = 1$. It will reappear in B.4.

B.1 Task 0.1 — Is the system reachable? Is it controllable?

Step 1: build $W_c = [\, B,\; AB,\; A^2B \,]$ (Concepts §5.1, $n = 3$, so we need up to A^2B).

$$B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 + 0 \\ 0 + 2 \cdot 0 + 2 \cdot 1 \\ 0 + 1 \cdot 0 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

$$A^2B = A(AB) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \cdot 2 + 2 \cdot 3 \\ 1 \cdot 2 + 3 \cdot 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \\ 11 \end{bmatrix}.$$

$$W_c = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 1 & 3 & 11 \end{bmatrix}.$$

Step 2: rank by row reduction.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 1 & 3 & 11 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 0 & 2 & 10 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(W_c) = 2 < 3 = n.$$

Cross-check with the determinant: $\det W_c = 1(2 \cdot 11 - 10 \cdot 3) - 1(0 \cdot 11 - 10 \cdot 1) + 1(0 \cdot 3 - 2 \cdot 1) = -8 + 10 - 2 = 0$ ✓.

The dependence is explicit: $A^2B = 5\,AB - 4\,B$, since $5[1,2,3] - 4[1,0,1] = [1,10,11]$. This is Cayley–Hamilton at work. With $p(\lambda) = \lambda^3 - 6\lambda^2 + 9\lambda - 4$, the minimal relation for this B happens to come from the factor $(\lambda - 1)(\lambda - 4) = \lambda^2 - 5\lambda + 4$.

Conclusion: the system is **not reachable** (not fully state controllable).

Step 3: PBH test — which eigenvalue is the problem? (Concepts §5.6)

For $\lambda = 4$:

$$[4I - A \quad B] = \begin{bmatrix} 3 & 0 & 0 & 1 \\ 0 & 2 & -2 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix}.$$

Row 1 is the only row with a nonzero first entry, so it is independent of the others. Rows 2 and 3 are not multiples of each other: $R_3 = -\frac{1}{2}R_2$ would need the last entry to be 0, but it is 1. So the rank is 3, and $\lambda = 4$ **is controllable**.

For $\lambda = 1$:

$$[I - A \quad B] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & -1 & -2 & 0 \\ 0 & -1 & -2 & 1 \end{bmatrix}.$$

$R_3 = R_1 + R_2$, so the rank is $2 < 3$. $\lambda = 1$ **is uncontrollable.**

Step 4: controllable in the sense of Def. 1.2? Controllability (steering every x_0 to the origin) would require every uncontrollable mode to vanish by itself in finite time, which needs eigenvalue 0. Here the uncontrollable eigenvalue is $\lambda = 1$. That mode neither grows nor decays, and the input cannot influence it.

> **Conclusion:** the system is **not controllable**. It is not even **stabilizable**: the uncontrollable eigenvalue 1 is not strictly inside the unit circle, so no K can make $A - BK$ asymptotically stable.

Step 5: intuition — a conserved quantity (Concepts §5.5). Take $w = [-1, -1, 1]^\top$. Then $w^\top A = -R_1 - R_2 + R_3 = [-1, -2 + 1, -2 + 3] = [-1, -1, 1] = 1 \cdot w^\top$, and $w^\top B = -1 - 0 + 1 = 0$. So $z_k := w^\top x_k = x_{3,k} - x_{1,k} - x_{2,k}$ satisfies

$$z_{k+1} = w^\top A x_k + w^\top B u_k = z_k + 0.$$

The quantity $x_3 - x_1 - x_2$ never changes, whatever input is applied. A system starting with $z_0 \neq 0$ can never reach the origin, where $z = 0$. That is exactly why it is not controllable.

Python check (`python/task0_reachability_observability.py`):

```
Eigenvalues of A: [4. 1. 1.]
Reachability matrix R = [B, AB, A^2 B]:
 [[ 1.  1.  1.]
 [ 0.  2. 10.]
 [ 1.  3. 11.]]
rank(R) = 2  (n = 3) -> NOT reachable
PBH test for each distinct eigenvalue:
  lambda = 1: rank[lambda I - A, B] = 2 -> uncontrollable mode
  lambda = 4: rank[lambda I - A, B] = 3 -> controllable mode
```

B.2 Task 0.2 — Reachable subspace and target states

Step 1: reachable subspace $=$ im(W_c) (Concepts §5.4). Since A^2B depends on the first two columns,

$$\mathcal{R} = \text{span}\left\{ \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\2\\3\end{bmatrix} \right\},$$

which is a plane through the origin (dimension 2 = rank).

Step 2: describe the plane with one equation using the cross product (Concepts §1.6):

$$\begin{bmatrix}1\\0\\1\end{bmatrix} \times \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}0\cdot 3 - 1\cdot 2\\1\cdot 1 - 1\cdot 3\\1\cdot 2 - 0\cdot 1\end{bmatrix} = \begin{bmatrix}-2\\-2\\2\end{bmatrix} \propto \begin{bmatrix}-1\\-1\\1\end{bmatrix}.$$

So the plane is $-x_1 - x_2 + x_3 = 0$:

$$\boxed{\mathcal{R} = \{\, x \in \mathbb{R}^3 : x_3 = x_1 + x_2 \,\}}$$

Check: B: $1 = 1 + 0$ ✓; AB: $3 = 1 + 2$ ✓; A^2B: $11 = 1 + 10$ ✓. This normal vector is exactly the conserved-quantity vector w from B.1 Step 5. Starting from $x_0 = 0$ we have $z = 0$ forever, so we stay in the plane $x_3 - x_1 - x_2 = 0$.

Step 3: target $x_f^{(1)} = [4, 2, 3]^\top$**.** $x_1 + x_2 = 4 + 2 = 6 \neq 3 = x_3$. The target is not in $\mathcal{R}$.

$x_f^{(1)}$ **is not reachable** from the origin, with any number of steps or any inputs.

(Rank test: rank$[W_c \;\; x_f^{(1)}] = 3 \neq 2$.)

Step 4: target $x_f^{(2)} = [6, 2, 8]^\top$**.** $x_1 + x_2 = 6 + 2 = 8 = x_3$ ✓. **Reachable.**

Step 5: minimum number of steps. From $x_0 = 0$, after N steps $x_N = \sum_{j=0}^{N-1} A^{N-1-j} B u_j$.

Try $N = 1$*:* $x_1 = Bu_0 = [u_0,\ 0,\ u_0]^\top$. The second component is always 0, but we need 2. **Impossible.**

Try $N = 2$*:* $x_2 = AB\,u_0 + B\,u_1$:

$$u_0 \begin{bmatrix}1\\2\\3\end{bmatrix} + u_1 \begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}6\\2\\8\end{bmatrix} \quad\Longleftrightarrow\quad \begin{cases} u_0 + u_1 = 6 \\ 2u_0 = 2 \\ 3u_0 + u_1 = 8 \end{cases}$$

- Row 2: $u_0 = 1$.
- Row 1: $u_1 = 6 - 1 = 5$.
- Row 3 (consistency check): $3 \cdot 1 + 5 = 8$ ✓.

$x_f^{(2)}$ **is reachable in a minimum of** $N = 2$ **steps, with** $u_0 = 1,\ u_1 = 5$**.** (The solution is unique, because B and AB are independent.)

Step 6: verify by simulation.

$$x_1 = A\cdot 0 + B\cdot 1 = \begin{bmatrix}1\\0\\1\end{bmatrix}, \qquad x_2 = Ax_1 + B\cdot 5 = \begin{bmatrix}1\\2\\3\end{bmatrix} + \begin{bmatrix}5\\0\\5\end{bmatrix} = \begin{bmatrix}6\\2\\8\end{bmatrix} \checkmark$$

Python check:

```
x_f1 = [4. 2. 3.]: reachable? False
x_f2 = [6. 2. 8.]: reachable? True
  minimum number of steps N = 2, inputs u = [1. 5.]
    x_1 = [1. 0. 1.]
    x_2 = [6. 2. 8.]
```

B.3 Task 0.3 — Is the system observable?

Step 1: build $W_o = [\,C;\ CA;\ CA^2\,]$ (Concepts §6.1). A row vector times A is a weighted sum of the rows of A.

$$C = [1 \quad 1 \quad 1]\,.$$

$$CA = 1\cdot[1,0,0] + 1\cdot[0,2,2] + 1\cdot[0,1,3] = [1 \quad 3 \quad 5]\,.$$

$$CA^2 = (CA)A = 1\cdot[1,0,0] + 3\cdot[0,2,2] + 5\cdot[0,1,3] = [1,\ 6+5,\ 6+15] = [1 \quad 11 \quad 21]\,.$$

$$W_o = \begin{bmatrix}1 & 1 & 1\\ 1 & 3 & 5\\ 1 & 11 & 21\end{bmatrix}.$$

Step 2: rank.

$$\xrightarrow{R_2-R_1,\ R_3-R_1} \begin{bmatrix}1 & 1 & 1\\ 0 & 2 & 4\\ 0 & 10 & 20\end{bmatrix} \xrightarrow{R_3-5R_2} \begin{bmatrix}1 & 1 & 1\\ 0 & 2 & 4\\ 0 & 0 & 0\end{bmatrix}$$

$$\text{rank}(W_o) = 2 < 3.$$

(Again $CA^2 = 5\,CA - 4\,C$: $5[1,3,5] - 4[1,1,1] = [1,11,21]$ ✓.)

Conclusion: the system is **not observable**.

Step 3: PBH test.

$\lambda = 4$: $\begin{bmatrix}4I - A\\ C\end{bmatrix} = \begin{bmatrix}3 & 0 & 0\\ 0 & 2 & -2\\ 0 & -1 & 1\\ 1 & 1 & 1\end{bmatrix}$. Rows 1, 2 and 4 give det $\begin{bmatrix}3 & 0 & 0\\ 0 & 2 & -2\\ 1 & 1 & 1\end{bmatrix} = 3(2\cdot 1 - (-2)\cdot 1) = 12 \neq 0$, so the rank is 3. **Observable mode.**

$\lambda = 1$: $\begin{bmatrix} I - A \\ C \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix}$. Rows 2 and 3 are equal and row 1 is zero, so only $[0, -1, -2]$ and $[1, 1, 1]$ are independent. The rank is $2 < 3$. **Unobservable mode.**

Equivalently, the $\lambda = 1$ eigenvector $v = [1, -2, 1]^\top$ has $Cv = 1 - 2 + 1 = 0$: the sensor cannot see it. The system is also **not detectable**, because the unobservable eigenvalue 1 is not inside the unit circle.

Python check:

```
Observability matrix O = [C; CA; CA^2]:
 [[ 1.  1.  1.]
 [ 1.  3.  5.]
 [ 1. 11. 21.]]
rank(O) = 2  (n = 3) -> NOT observable
  lambda = 1: rank[lambda I - A; C] = 2 -> unobservable mode
  lambda = 4: rank[lambda I - A; C] = 3 -> observable mode
```

B.4 Task 0.4 — Observable subspace and initial states

Step 1: unobservable subspace $= \ker(W_o)$**.** Solve $W_o v = 0$. Use the two independent rows:

$$v_1 + v_2 + v_3 = 0 \qquad (1)$$
$$v_1 + 3v_2 + 5v_3 = 0 \qquad (2)$$

$(2) - (1)$: $2v_2 + 4v_3 = 0 \Rightarrow v_2 = -2v_3$. Into (1): $v_1 = -v_2 - v_3 = 2v_3 - v_3 = v_3$. With $v_3 = t$: $v = t\,[1, -2, 1]^\top$.

Check with row 3: $1 - 22 + 21 = 0$ ✓.

$$\boxed{\ker(W_o) = \text{span}\{[1,\ -2,\ 1]^\top\}} \quad \text{(a line, dimension } 3 - 2 = 1\text{)}$$

Step 2: observable subspace $= \ker(W_o)^\perp$ = row space of W_o:

$$\boxed{\mathcal{O} = \{\, x : x_1 - 2x_2 + x_3 = 0 \,\} = \text{span}\{[1, 1, 1]^\top,\ [1, 3, 5]^\top\}}$$

Check: $[1, 1, 1]$: $1 - 2 + 1 = 0$ ✓; $[1, 3, 5]$: $1 - 6 + 5 = 0$ ✓.

Step 3: initial state $x_0^{(1)} = [1, 2, 3]^\top$**.**

- Test for membership in $\mathcal{O}$: $1 - 2 \cdot 2 + 3 = 0$ ✓. So $x_0^{(1)}$ lies in the observable subspace, and its unobservable part is zero.
- Its outputs: $\vec{y} = W_o x_0^{(1)} = [1 + 2 + 3,\ 1 + 6 + 15,\ 1 + 22 + 63]^\top = [6,\ 22,\ 86]^\top$, which is nonzero.

  $x_0^{(1)}$ **can be observed**: it is the unique state in the observable subspace that produces the outputs $6, 22, 86$. *Fine print:* since the system is not observable, any $x_0^{(1)} + c\,[1, -2, 1]^\top$ would produce the same outputs. The measurements identify the observable component, and $x_0^{(1)}$ consists only of that component.

Step 4: initial state $x_0^{(2)} = [-2, 4, -2]^\top$.

- $x_0^{(2)} = -2\,[1, -2, 1]^\top$, so it lies entirely in $\ker(W_o)$.
- Outputs: $y_0 = -2 + 4 - 2 = 0$; $y_1 = -2 + 12 - 10 = 0$; $y_2 = -2 + 44 - 42 = 0$.
- Because $[1, -2, 1]$ is an eigenvector for $\lambda = 1$, the state never moves ($x_k = x_0^{(2)}$ for all k), and $Cx_k = 0$ forever.

  $x_0^{(2)}$ **cannot be observed**: it produces $y_k = 0$ for all k, exactly like $x_0 = 0$.

Python check:

```
Basis of UNobservable subspace (kernel of O): [-0.4082  0.8165 -0.4082]   (= multiple of [1,
x0_1 = [1. 2. 3.]
  output sequence y_0..y_2 = [ 6. 22. 86.]
  -> lies in the observable subspace: can be observed
x0_2 = [-2.  4. -2.]
  output sequence y_0..y_2 = [0. 0. 0.]
  -> lies entirely in the unobservable subspace: cannot be observed
```

Task 0 — summary

Question	Answer
Eigenvalues of A	$1, 1, 4$
rank W_c	2 — **not reachable**
Uncontrollable mode (PBH)	$\lambda = 1$ — **not controllable**, not stabilizable
Reachable subspace	plane $x_3 = x_1 + x_2$
$x_f^{(1)} = [4, 2, 3]$	not reachable
$x_f^{(2)} = [6, 2, 8]$	reachable, min. $N = 2$, $(u_0, u_1) = (1, 5)$
rank W_o	2 — **not observable**
Unobservable mode (PBH)	$\lambda = 1$ — not detectable
Unobservable subspace	span$\{[1, -2, 1]\}$
Observable subspace	plane $x_1 - 2x_2 + x_3 = 0$
$x_0^{(1)} = [1, 2, 3]$	observable ($y = 6, 22, 86$)
$x_0^{(2)} = [-2, 4, -2]$	not observable ($y \equiv 0$)

Part C — Task 1: Tracking error for a state-feedback controller

Setting: $x_{k+1} = Ax_k + Bu_k$, controller $u_k = -K(x_k - x_{\text{ref}})$ with constant x_{ref}, and K such that $A - BK$ is asymptotically stable.

C.0 Setting up the closed loop

Substitute the controller into the system:

$$\begin{aligned} x_{k+1} &= Ax_k + B(-K(x_k - x_{\text{ref}})) \\ &= Ax_k - BKx_k + BKx_{\text{ref}} \\ &= \underbrace{(A - BK)}_{=:M} x_k + BKx_{\text{ref}}. \end{aligned}$$

So the closed loop is $x_{k+1} = Mx_k + c$ with constant $c = BKx_{\text{ref}}$, and by assumption all eigenvalues of M satisfy $|\lambda| < 1$.

C.1 Task 1.1 — $I - (A - BK)$ is invertible and $x_k \to x_\infty$

Part (a): invertibility of $I - M$ Proof by contradiction.

1. Suppose $I - M$ is **not** invertible.
2. Then there is a vector $v \neq 0$ with $(I - M)v = 0$ (Concepts §1.7).
3. Expand: $v - Mv = 0$, i.e. $Mv = v = 1 \cdot v$.
4. So v is an eigenvector of M with eigenvalue $\lambda = 1$.
5. But M is asymptotically stable, so every eigenvalue satisfies $|\lambda| < 1$. Since $|1| = 1$, this is a contradiction.
6. Therefore $I - M = I - (A - BK)$ **is invertible**. ■

   Remember: $I - M$ is singular $\iff$ 1 is an eigenvalue of M.

Part (b): convergence to $x_\infty = (I - (A - BK))^{-1} BKx_{\text{ref}}$ Step 1: find the equilibrium. A limit must be an equilibrium (Concepts §4.3), so set $x_{k+1} = x_k = x_\infty$:

$$x_\infty = Mx_\infty + BKx_{\text{ref}} \implies x_\infty - Mx_\infty = BKx_{\text{ref}} \implies (I - M)\, x_\infty = BKx_{\text{ref}}.$$

By part (a), $I - M$ is invertible, so there is exactly one equilibrium:

$$x_\infty = (I - M)^{-1} BKx_{\text{ref}} = \big(I - (A - BK)\big)^{-1} BKx_{\text{ref}}.$$

Step 2: show that the trajectory actually converges to it (shift trick, Concepts §4.4). Define $d_k := x_k - x_\infty$:

$$\begin{aligned} d_{k+1} &= x_{k+1} - x_\infty \\ &= \big(Mx_k + BKx_{\text{ref}}\big) - \big(Mx_\infty + BKx_{\text{ref}}\big) \quad \text{(both satisfy the same update)} \\ &= M(x_k - x_\infty) \\ &= Md_k. \end{aligned}$$

By induction, $d_k = M^k d_0$. Since M is asymptotically stable, $M^k \to 0$, hence $d_k \to 0$ for **every** x_0:

$$\lim_{k\to\infty} x_k = x_\infty = \big(I - (A - BK)\big)^{-1} BKx_{\text{ref}}. \qquad \blacksquare$$

Alternative proof (via the lecture's Proposition 1.1) The closed loop is a system with matrix M and constant "input" BKx_{ref}. By the solution formula,

$$x_k = M^k x_0 + \sum_{j=0}^{k-1} M^{k-1-j} BKx_{\text{ref}} = M^k x_0 + \Big(\sum_{i=0}^{k-1} M^i\Big) BKx_{\text{ref}}.$$

As $k \to \infty$: $M^k x_0 \to 0$, and by the geometric series (Concepts §4.2) $\sum_{i=0}^{\infty} M^i = (I - M)^{-1}$. This gives the same x_∞.

C.2 Task 1.2 — Asymptotic tracking error and when it can be zero

Step 1: the one-dimensional case (as the hint suggests) Let $A = a$, $B = b$, $K = k$, all scalars. The closed loop is $x_{k+1} = (a - bk)x_k + bk\, x_{\text{ref}}$.

- **Stability:** $|a - bk| < 1$.
- **Limit:** from C.1, $x_\infty = \dfrac{bk}{1-(a-bk)}\, x_{\text{ref}} = \dfrac{bk}{1-a+bk}\, x_{\text{ref}}$.
- **Error:** put both terms over a common denominator:

$$\begin{aligned} x_\infty - x_{\text{ref}} &= \frac{bk}{1-a+bk} x_{\text{ref}} - \frac{1-a+bk}{1-a+bk} x_{\text{ref}} \\ &= \frac{bk-1+a-bk}{1-a+bk}\, x_{\text{ref}} \\ &= \boxed{\frac{a-1}{1-a+bk}\, x_{\text{ref}}} \end{aligned}$$

Observation: the numerator $a - 1$ **does not contain** k. Therefore:

- If $a = 1$: the error is 0 for every stabilizing k ✓.
- If $a \neq 1$ (and $x_{\text{ref}} \neq 0$): the error is **never** zero, whatever k is.

Can the error at least be made small? A larger bk makes the denominator larger, but stability restricts it: $|a - bk| < 1 \iff a - 1 < bk < a + 1$. So $1 - a + bk < 2$, and the error is bounded below in size by $\frac{|a-1|}{2}|x_{\text{ref}}|$. It cannot be made arbitrarily small.

Numerical example: $a = 0.5$, $b = 1$. Stability requires $bk \in (-0.5,\, 1.5)$.

k	$a - bk$ (closed-loop pole)	$x_\infty / x_{\text{ref}} = \frac{bk}{1-a+bk}$	error $/x_{\text{ref}}$
0.5	0	$0.5/1.0 = 0.5$	-0.5
1.0	-0.5	$1.0/1.5 \approx 0.667$	≈ -0.333
1.4	-0.9	$1.4/1.9 \approx 0.737$	≈ -0.263
$\to 1.5$	$\to -1$ (stability limit)	$\to 0.75$	$\to -0.25$

You never get closer than 75 % of the way to the target.

Step 2: the general (matrix) case Use the trick $x_{\text{ref}} = (I - M)^{-1}(I - M)\, x_{\text{ref}}$, which is valid because $I - M$ is invertible:

$$\begin{aligned} x_\infty - x_{\text{ref}} &= (I-M)^{-1}BKx_{\text{ref}} - (I-M)^{-1}(I-M)x_{\text{ref}} \\ &= (I-M)^{-1}\big[BK - (I-M)\big]x_{\text{ref}} \\ &= (I-M)^{-1}\big[BK - I + A - BK\big]x_{\text{ref}} \qquad (M = A - BK) \\ &= (I-M)^{-1}(A-I)\, x_{\text{ref}}. \end{aligned}$$

$$\boxed{x_\infty - x_{\text{ref}} = \big(I - A + BK\big)^{-1}(A-I)\, x_{\text{ref}}}$$

For $n = 1$ this reduces to the scalar formula above.

Step 3: when is the error zero? $(I - A + BK)^{-1}$ is invertible, so it maps only the zero vector to zero. Therefore

$$x_\infty - x_{\text{ref}} = 0 \iff (A - I)\, x_{\text{ref}} = 0 \iff A\, x_{\text{ref}} = x_{\text{ref}}.$$

Answer to Task 1.2. The asymptotic tracking error is $x_\infty - x_{\text{ref}} = (I - A + BK)^{-1}(A - I)x_{\text{ref}}$. It is zero **if and only if** $Ax_{\text{ref}} = x_{\text{ref}}$, i.e. x_{ref} is already an equilibrium of the *uncontrolled* system ($u = 0$), or equivalently $x_{\text{ref}} \in \ker(A - I)$ (an eigenvector of A for eigenvalue 1, or $x_{\text{ref}} = 0$). - If this holds, **any** stabilizing K achieves $x_\infty = x_{\text{ref}}$. - If it does not hold, **no** choice of K achieves zero error. The condition does not depend on K at all. - Zero error for **every** reference requires $A = I$ (in 1-D: $a = 1$).

Step 4: interpretation (why this happens) To hold the state at x_{ref}, the input must satisfy $x_{\text{ref}} = Ax_{\text{ref}} + B\bar{u}$, i.e. $B\bar{u} = (I - A)x_{\text{ref}}$. This input is nonzero unless $Ax_{\text{ref}} = x_{\text{ref}}$. But at $x = x_{\text{ref}}$ the P-controller outputs $u = -K \cdot 0 = 0$. The controller can only produce input by keeping an error, so the system settles where the error is just large enough to supply the needed input. This is the classic **steady-state error of a P-controller** (Concepts §7.3).

This motivates Task 2 (rescale the reference with a prefilter F) and Task 3 (add an integrator, i.e. a PI controller).

Step 5: numerical verification (`python/task1_tracking_error.py`)

Case	Simulated x_∞	Formula
1-D: $a = 0.5,\ b = 1,\ k = 0.5,\ x_{\text{ref}} = 1$	0.5	$\frac{0.5}{1.0} = 0.5$ ✓
System a) of the numerical experiment, $K = \begin{bmatrix} 0 & 0.2 \\ 0.6 & 0 \end{bmatrix}$, $x_{\text{ref}} = [1, 0]^\top$	$[0.931,\ 0.103]$	$[0.931,\ 0.103]$ ✓
Error for the same case	$[-0.069,\ 0.103]$	$(I - A + BK)^{-1}(A - I)x_{\text{ref}} = [-0.069,\ 0.103]$ ✓

For system a), the closed-loop eigenvalues are 0.445 and -0.045 (stable), but $A[1, 0]^\top = [0.9, 0.1]^\top \neq [1, 0]^\top$, so a steady-state error remains, as the theory predicts.